DiMA-PPI: Canonical State-Potential Learning for Consistent Protein Binding Mutation Prediction

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***Abstract*—Predicting mutation-induced changes in protein–protein binding free energy supports variant interpretation and protein engineering. Independently predicted substitution effects can produce path-dependent estimates of the binding free-energy difference between the same protein variants. We propose DiMA-PPI, a multimodal graph neural network that enforces path consistency through shared state-potential learning. The model integrates sequence embeddings, structural representations and local atomic descriptors to estimate latent potentials for bound and isolated-chain states. Mutation effects are expressed as differences between the binding potentials of the initial and final variants. Canonical state reuse maintains a single numerical value for each represented state across all predictions. On S1129, comprising 1,129 substitutions from 111 complexes, DiMA-PPI achieves a Pearson correlation coefficient (PCC) of 0.757 and a root mean squared error (RMSE) of 1.542 kcal mol^{-1}. We further construct PPI-Path92, which contains 92 groups of same-site variants and 9,904 mutant-to-mutant comparisons whose targets are derived from measured effects relative to the wild type. On PPI-Path92, DiMA-PPI achieves an RMSE of 1.611 kcal mol^{-1} and a mean discrepancy of 4.78×10^{-7} kcal mol^{-1} between direct and multistep predictions. Experiments with fixed model parameters show that state reuse eliminates inconsistencies arising from independent stochastic evaluation and reduces redundant computation. These results support canonical state-potential learning for consistent prediction of binding changes across related protein variants.**

***Index Terms*—Graph neural networks, state potentials, protein binding, path consistency.**

I. Introduction

Predicting changes in protein–protein binding free energy following amino-acid substitution is a central task in variant interpretation and protein engineering. Established tools include FoldX [1], MutaBind2 [2], mCSM-PPI2 [3] and DDMut-PPI [4]. Graph neural networks and pretrained protein representations integrate structural and sequence information to estimate these effects [5], [6]. Evaluation on benchmarks such as SKEMPI [7] primarily measures accuracy for individual substitutions. Comparison of related variants additionally requires path consistency: for states a, b and c under the same modeled conditions, the predicted change from a to c should equal the sum of the changes from a to b and from b to c. This relation couples predictions across substitutions.

Reverse-mutation augmentation and antisymmetric prediction address directional bias [8] by relating each substitution to its reverse. Global consistency additionally requires zero accumulated change around closed mutation cycles. Representing protein states as graph nodes and mutation effects as directed edges provides a unified formulation: edge values defined by node-potential differences sum to zero around every cycle, analogous to voltage differences around a circuit loop. Implementing this property requires identical numerical representations of shared states throughout structure preparation and inference.

StaB-ddG derives binding effects from folding scores [9], while DSSA-PPI uses residue-score differences under a shared conditioning context [6]. We introduce canonical state identity across structure preparation, feature extraction and inference to preserve potential differences throughout the prediction pipeline. DiMA-PPI learns a binding potential from each pair of bound and free representations and reuses canonical state values across mutation queries. We also construct PPI-Path92 from S1129 to evaluate compositional prediction and cycle consistency. Matched model comparisons assess the relationship between predictive accuracy and consistency, while experiments with fixed model parameters quantify the numerical and computational effects of state reuse.

II. Proposed Method

Figure 1 summarizes the shared encoder, four-state output and canonical inference procedure.

A. Canonical Bound and Free States

For a fixed complex and substitution site, S_a denotes the state with amino-acid identity a. Positive $y_{a\to b} = \Delta\Delta G_{a\to b}$ follows the benchmark convention of stronger mutant binding. Each state has a bound context S_a^b and an isolated-chain free context S_a^f, identified by

$$k(a,c) = (\text{complex}, \text{chain}, \text{PDB site}, a, c), \quad c \in \{b, f\}. \quad (1)$$

Each key maps to one representation selected by source-row order, independently of labels and fold assignments. Repeated occurrences reuse identical sequence features, graph geometry, node masks and atomic descriptors, ensuring a consistent representation of shared intermediate states.

The reference-backbone graph contains residues within 20 Å of the site. Edges connect residues within 8 Å and sequential neighbors on each chain; 16 radial basis functions encode their distances. The bound encoder uses the complex graph. For the free context, cross-chain edges are removed and readout is restricted to the mutated chain.

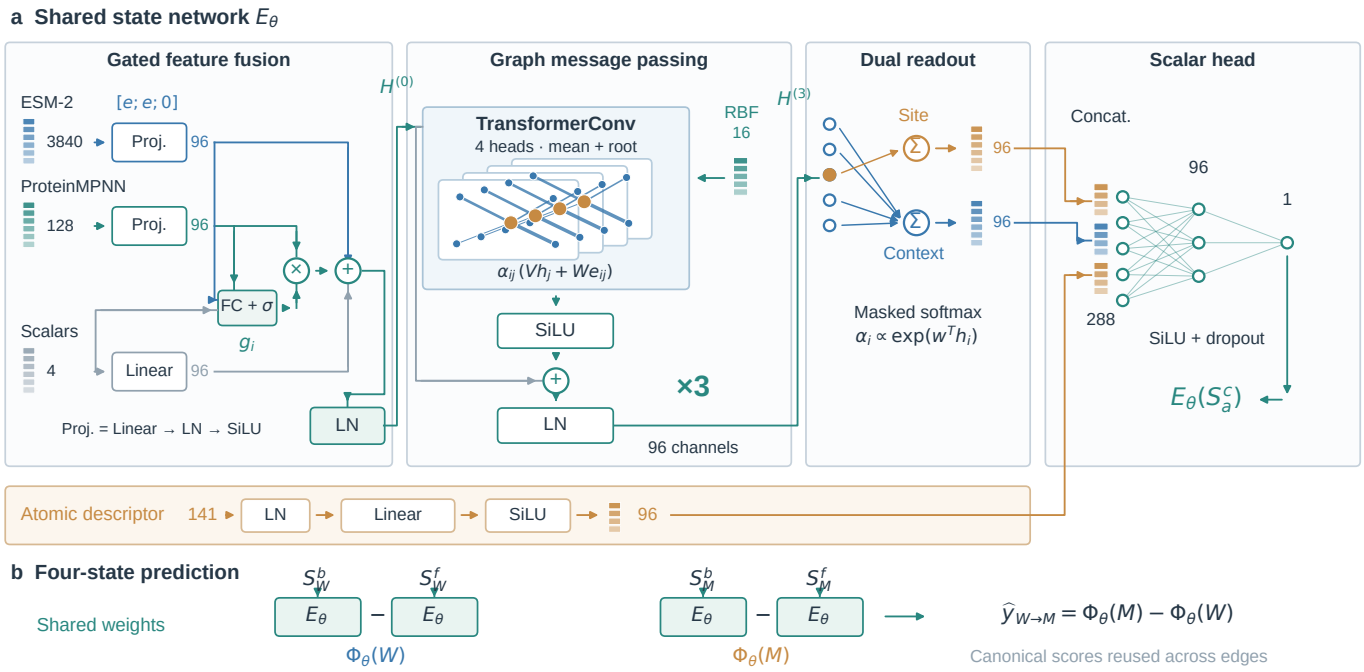


Fig. 1. DiMA-PPI network architecture. (a) Projected sequence and backbone features enter a learned sigmoid gate; elementwise multiplication and addition form the fused node features. Each of three graph blocks contains four-head TransformerConv, SiLU, a residual addition and LayerNorm (LN). Edge RBFs enter attention; line widths schematically indicate attention weights. Masked attention pools the mutation site and context separately. Their two 96-D vectors are concatenated with a projected 96-D atomic descriptor and passed through a 288–96–1 scalar head. Drawn neurons represent layers rather than exact neuron counts. (b) The same network scores all four bound/free and WT/mutant states. Bound-minus-free scores define binding potentials, whose difference gives the mutation effect. Canonical scores are reused across edges.

Four 141-dimensional descriptors encode wild-type/mutant and bound/free atomic environments. PyRosetta [10] locally repacks side chains within 8 Å in the bound structures and repacks the isolated mutated chain for the free contexts. Descriptors contain element-distance histograms, substituted-residue element counts, 18 residue-level Rosetta score terms and their local sums.

B. Shared Multimodal Graph Encoder

Each residue has an ESM-2 650M embedding $e_i^a \in \mathbb{R}^{1280}$ [11], a ProteinMPNN reference-backbone feature $p_i \in \mathbb{R}^{128}$ [12], and four scalars s_i encoding chain membership, site membership and normalized site distance. The single-state sequence block is $u_i^a = [e_i^a; e_i^a; \mathbf{0}]$, which represents a single sequence identity in both embedding blocks and fixes the difference block to zero. A learned gate integrates the sequence and structural features:

$$\begin{aligned} q_i &= \phi_{\text{seq}}(u_i^a), \quad t_i = \phi_{\text{str}}(p_i), \\ g_i &= \sigma(W_g[q_i; t_i; s_i] + b_g), \\ h_i^{(0)} &= \text{LN}(q_i + g_i \odot t_i + W_s s_i + b_s). \end{aligned} \tag{2}$$

The projections use linear layers, layer normalization and SiLU. Three TransformerConv layers with four averaged attention heads and hidden width 96 propagate messages over the context-specific edges:

$$H^{(\ell+1)} = \text{LN}\Big[H^{(\ell)} + \text{SiLU}\left(\text{TConv}_\ell(H^{(\ell)}, \mathcal{E}_c, R)\right)\Big]. \tag{3}$$

Attention pooling produces separate representations of the substitution site and surrounding context, with weights proportional to $\exp(w^\top h_i^{(3)})$ within each node set. A shared scalar head maps these representations and the atomic descriptor to a state score:

$$E_\theta(S_a^c) = \text{MLP}_\theta([z_{\text{site}}; z_{\text{context}}; \phi_d(d_a^c)]), \tag{4}$$

where ϕ_d normalizes and projects the atomic descriptor to width 96. All four evaluations share encoder, projection and scalar-head weights.

C. Potential Differences and Numerical Identity

Define the latent binding potential and mutation prediction as

$$\Phi_\theta(a) = E_\theta(S_a^b) - E_\theta(S_a^f), \tag{5}$$

$$\hat{y}_{a\to b} = \Phi_\theta(b) - \Phi_\theta(a). \tag{6}$$

For any path, intermediate potentials cancel:

$$\sum_{\ell=0}^{L-1} \hat{y}_{a_\ell \to a_{\ell+1}} = \Phi_\theta(a_L) - \Phi_\theta(a_0). \tag{7}$$

Equation (7) establishes antisymmetry, path independence and zero cycle sums. For an oriented node–edge incidence matrix B, the prediction vector is $B^\top \mathbf{\Phi}$, a discrete potential-difference field with zero circulation. In comparison, pairwise

antisymmetrization $[f(a,b)-f(b,a)]/2$ constrains the reversal relation.

Canonical inference preserves the state identity required by Eq. (7). With dropout disabled, each context score can be computed once per checkpoint and reused across all incident edges. For stochastic inference, the same state estimate is reused within each draw. Both procedures preserve cancellation to floating-point precision. Mutation-effect supervision determines potential differences; a common additive offset leaves every prediction invariant. The represented state space is defined by the complex, substitution site and structural context.

D. Optimization

The model is optimized using observed forward mutation effects. With the training-subset target standard deviation σ_y and scaled four-state output $\widetilde{y}_\theta$, the objective is

$$\mathcal{L} = \frac{1}{N}\sum_n \mathrm{SmoothL1}_{0.5}(\widetilde{y}_\theta(a_n,b_n) - y_n/\sigma_y). \quad (8)$$

Predictions are rescaled to physical units by multiplication with σ_y, with the additive offset fixed at zero. The objective consists solely of the forward-prediction loss; consistency is imposed by the output parameterization. AdamW uses learning rate 5×10^{-4}, weight decay 10^{-4}, batch size 8 and dropout 0.15. Training runs for up to 60 epochs with validation-MSE early stopping and patience 10.

III. Experiments and Results

A. Datasets and Evaluation Protocol

S1129. The public benchmark contains 1,129 single substitutions from 111 complexes following the two official exclusions from S1131. We use the five predefined complex-disjoint outer folds from the DSSA-PPI protocol [6], with test sizes 169, 142, 271, 252 and 295. Approximately one eighth of each outer training fold is reserved for validation using a seeded mutation-level split; complexes may therefore occur in both fitting and validation subsets. We report the mean and sample SD across seeds 42–46 after averaging per-fold metrics within each seed. PCC, Spearman correlation and RMSE measure linear association, rank agreement and error magnitude, respectively.

PPI-Path92. We construct the Protein–Protein Interaction Mutation Path Benchmark (PPI-Path92) from repeated-site measurements in S1129. Measurements are grouped by complex, chain, PDB site and WT residue. Sites with at least two distinct measured mutant variants are retained, with duplicate records resolved by selecting the earliest source row. The resulting benchmark contains 92 site panels from 27 complexes and 793 amino-acid states. For each ordered mutant pair, the target is

$$y^{\mathrm{derived}}_{b\to c} = y_{\mathrm{WT}\to c} - y_{\mathrm{WT}\to b}. \quad (9)$$

PPI-Path92 contains 9,904 mutant-to-mutant comparisons with targets defined by Eq. (9). Native predictions evaluate each mutant pair directly; endpoint composition subtracts the corresponding predicted WT effects. Including WT edges, the state graphs contain 11,306 directed edges, supporting 5,653 reverse comparisons, 9,904 ordered paths, 31,922 triangles and 133,736 four-state cycles per seed. Each panel inherits the S1129 outer fold of its source complex, preserving complex-disjoint evaluation.

TABLE I
S1129 predictive accuracy: mean ± sample SD across five repeats or seeds, each averaged over five folds. RMSE is in kcal mol^{-1}.

Model	PCC ↑	Spearman ↑	RMSE ↓
Published-model comparison			
DSSA-PPI [6]	0.706 ± 0.011	0.508 ± 0.017	1.717 ± 0.040
DiMA-PPI (ours)	0.757 ± 0.009	0.592 ± 0.011	1.542 ± 0.023
Matched controls and ablation			
Pairwise	0.570 ± 0.020	0.514 ± 0.021	2.112 ± 0.047
Reverse supervision	0.756 ± 0.009	0.602 ± 0.011	1.565 ± 0.039
Soft consistency	0.759 ± 0.012	0.608 ± 0.014	1.551 ± 0.045
Hard antisymmetry	0.755 ± 0.003	0.594 ± 0.009	1.549 ± 0.013
No state descriptors	0.750 ± 0.004	0.578 ± 0.006	1.588 ± 0.025

Matched controls share residue-graph inputs and training settings: an unconstrained pairwise regressor, reverse-label supervision, a soft reverse consistency penalty (weight 0.20), hard pairwise antisymmetry, and a shared potential without atomic state descriptors. Reverse and path residuals are $|\widehat{y}_{a\to b}+\widehat{y}_{b\to a}|$ and $|\widehat{y}_{a\to c}-\widehat{y}_{a\to b}-\widehat{y}_{b\to c}|$. Cycle residuals are absolute sums around the loop.

B. Comparison with Published Models and Ablation

DSSA-PPI [6] is evaluated using five released repeats of fold-specific checkpoints, using the same S1129 observations and outer folds as DiMA-PPI and retaining its original pretraining and optimization procedures.

DiMA-PPI achieves PCC 0.757 ± 0.009, Spearman correlation 0.592 ± 0.011 and RMSE 1.542 ± 0.023 kcal mol^{-1} (Table I). Relative to DSSA-PPI, DiMA-PPI increases mean PCC by 0.051 and Spearman correlation by 0.084, and reduces RMSE by 0.176 kcal mol^{-1}. Among matched controls, soft consistency achieves the highest mean PCC and Spearman correlation; DiMA-PPI achieves the lowest mean RMSE.

Removing state descriptors yields PCC 0.750, Spearman correlation 0.578 and RMSE 1.588. Paired uncertainty is estimated from five-seed ensemble predictions using 10,000 complex-bootstrap replicates within each fold. Including the descriptors increases ensemble PCC by 0.0098 (95% CI 0.0002–0.0206) and reduces RMSE by 0.0490 kcal mol^{-1} (95% CI 0.0138–0.0779).

C. Path and Cycle Consistency on PPI-Path92

Table II reports residuals for reversal, path composition and cycle closure. Hard antisymmetry achieves numerical precision for reversal, with mean path and four-cycle residuals of 0.441 and 0.348 kcal mol^{-1}. DiMA-PPI achieves mean residuals below 5×10^{-7} for all four relations. This evaluation recomputes scores from deterministic canonical inputs for each requested edge; the resulting residuals are consistent with floating-point arithmetic.

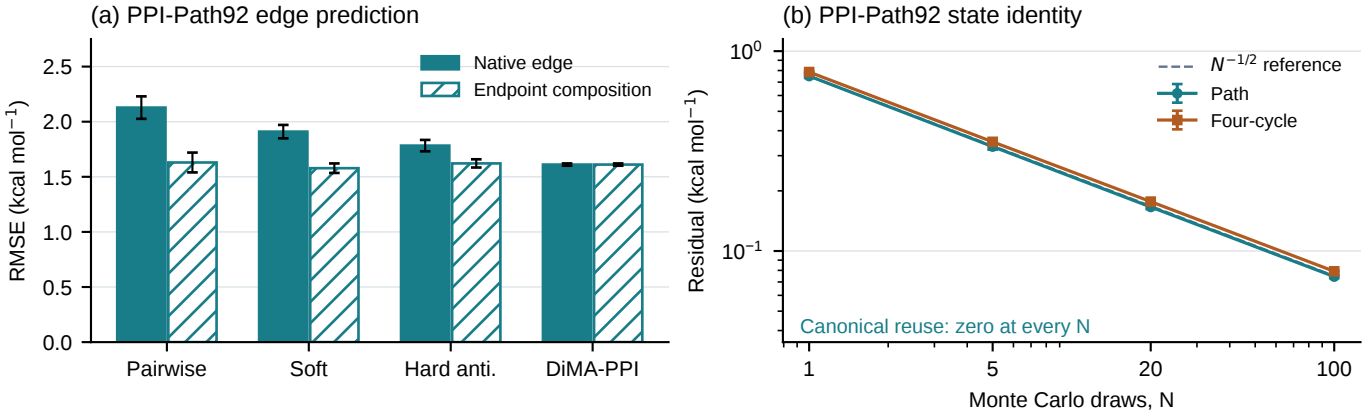


Fig. 2. Evaluation on PPI-Path92. (a) RMSE of native mutant-to-mutant predictions and composition of predicted WT effects, evaluated against differences of measured WT effects. Bars and error bars denote the mean and sample SD across five seeds. (b) Consistency residuals under independent state sampling with fixed StaB-ddG parameters. Canonical reuse yields zero measured residuals at all sample counts, as indicated below the curves.

TABLE II
PPI-PATH92 MEAN ABSOLUTE CONSISTENCY RESIDUALS (KCAL MOL^{-1}), AVERAGED ACROSS FIVE SEEDS.

Model	Reverse	Path	Triangle	Four-cycle
Pairwise	2.260	1.298	3.283	4.248
Soft	0.867	0.879	1.314	1.652
Hard anti.	1.63×10^{-7}	0.441	0.305	0.348
DiMA-PPI	2.20×10^{-7}	4.78×10^{-7}	3.08×10^{-7}	3.68×10^{-7}

D. Compositional Prediction on PPI-Path92

On the 9,904 mutant-to-mutant edges defined by Eq. (9), native pairwise, soft, hard and DiMA-PPI predictors achieve PCC 0.354, 0.442, 0.502 and 0.547, and RMSE 2.128, 1.910, 1.783 and 1.611 kcal mol^{-1}, respectively (Fig. 2a). The paired bootstrap of five-seed ensemble predictions estimates an RMSE reduction of 0.167 (95% CI 0.016–0.466) relative to hard antisymmetry. The ensemble PCC gain is 0.049 (95% CI -0.041 to 0.175).

Composition of predicted WT effects yields RMSE 1.630, 1.578 and 1.622 for the pairwise, soft and hard controls, respectively. Endpoint composition therefore recovers much of the accuracy difference observed for native predictions; the composed soft control has the lowest mean RMSE. In DiMA-PPI, native prediction and endpoint composition are algebraically equivalent through the shared state potential.

E. Effect of Canonical State Reuse

We assess the effect of state reuse with 25 fixed StaB-ddG checkpoints covering five folds and five seeds. Independent and canonical inference use the same panels and nested Monte Carlo sample counts $N \in \{1, 5, 20, 100\}$. Independent mean path error decreases from 0.751 to 0.074 kcal mol^{-1}, approximately following $N^{-1/2}$ (Fig. 2b). Reusing each state estimate across edges yields zero measured reverse, path and cycle residuals at every N. In the paired evaluation at $N = 20$, canonical reuse preserves the WT-to-mutant estimates and their predictive metrics while enforcing closure across the state graph.

A computational evaluation using the seed-42 DiMA-PPI checkpoints covers 793 amino-acid states, 1,586 bound/free contexts and 11,306 directed edges in PPI-Path92. Independent edge evaluation requests 45,224 context scores; canonical reuse requires 1,586, a 28.5-fold reduction. On an Intel Core i7-12700KF with eight PyTorch CPU threads and batch size 32, model inference time decreases from 469.4 to 16.1 seconds, corresponding to a 29.1-fold speedup. Checkpoint/cache loading, graph assembly and disk I/O are performed outside the timed region. The mean and maximum absolute prediction differences between the two inference procedures are 1.10×10^{-7} and 1.67×10^{-6} kcal mol^{-1}, and reused scores give zero measured closure residuals.

IV. Discussion and Conclusion

DiMA-PPI combines a shared four-state encoder with canonical state identity to produce globally consistent mutation predictions. It achieves PCC 0.757 on S1129 and RMSE 1.611 on the PPI-Path92 compositional benchmark, with path and cycle residuals at numerical precision. Fixed-weight experiments establish canonical scalar reuse as an effective way to preserve consistency and reduce repeated inference across state graphs.

Extensions to condition-aware states, conformational ensembles and multi-site mutation paths provide further directions for evaluation.

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